

CSC 309

Data Structure and Algorithm

Examination: C.A. Test	Semester: Second Semester
Year: 2024	Duration: 45 minutes
Total Questions: 3	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

Explain the following sorting algorithms with examples: (a) Bubble Sort (b) Selection Sort (c) Insertion Sort. State their time complexities. (12 marks)

Question 2

- (a) What is a binary search tree (BST)? (3 marks)
- (b) Construct a BST by inserting the following values in order: 50, 30, 70, 20, 40, 60, 80. (5 marks)
- (c) Write the in-order, pre-order, and post-order traversals of the resulting tree. (5 marks)

Question 3

Explain the concept of hashing. What are collision resolution techniques? Describe chaining and open addressing. (10 marks)

— End of Examination —